



Generate Censored Samples: Controlling censorship percentage

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Abstract

In this research [1], we provide a simple and general guide for researchers aiming to generate pseudo-random samples from censored data. We focus on the most common types of censoring: Type I, Type II, and Type III (random censoring). We describe the necessary steps to sample pseudo-random values from a general distribution while controlling the desired percentage of censoring. To illustrate these techniques, we apply them to the Weibull distribution, which is commonly used in survival analysis. Finally, we developed an R package that implements these methodologies, offering researchers practical and efficient tools for use in simulation studies.

Censorship Mechanisms

Right Censoring: This is the most common type of censoring, where the time of an event is observed up to a fixed time. Observations exceeding this time are considered censored. If T is the survival time and t_c is the censoring time, the relationship can be expressed as:

$$\text{Observed} = \min(T, t_c).$$

Left Censoring: In this case, it is known that an event (T) occurred before a specific time t_c , but it cannot be determined when.

$$\text{Observed} = \max(T, t_c).$$

Interval Censoring: This type of censoring occurs when there is a time range (interval) in which the event may have occurred, but the exact moment is unknown.

$$T \in [t_L, t_U].$$

Sample : Interval Censoring

We aim to simulate data with interval censoring based on a response variable T and a random variable C , which represents the time between observations when we can check whether T has occurred. We assume $T \perp C$. The goal is to generate an interval $I = [L, U]$ such that $T \in I$, where $L = \sum_{i=1}^n C_i$ and $U = \sum_{i=1}^{n+1} C_i$, with an appropriate $n \in \mathbb{N}$. We can control n by setting a lower bound such that:

$$P\left(\sum_{i=1}^n C_i \geq t\right) \geq \epsilon(n).$$

Let $Z = \sum_{i=1}^n C_i$, so that $E[Z] = n E[C]$ and $\text{Var}[Z] = n \text{Var}[C]$.

$$P(Z > t) \geq \frac{\left(n - \frac{t}{E[C]}\right)^2 E[C]^2}{n \text{Var}[C] + n^2 E[C]^2} = \epsilon(n).$$

This result, based on Paley–Zygmund inequality, provides a lower bound on the probability that the sum of the variables C_i exceeds the value t , as a function of n and the characteristics of C .

References

- [1] Pedro L. Ramos et al. “Sampling with censored data: a practical guide”. In: *Journal of Statistical Computation and Simulation* (Oct. 2024), pp. 1–35. DOI: 10.1080/00949655.2024.2409379.
- [2] Daniel Saavedra and Pedro L Ramos. *rcens: Generate Sample Censoring*. Mar. 2024. DOI: 10.32614/cran.package.rcens.

Type of Censoring

Type I censoring occurs when experiments are concluded at a predetermined time t_c . Observations that exceed this time are classified as censored, ensuring that results remain up-to-date.

Type II censoring involves stopping the experiment after a predetermined number of events have occurred. Given a total of n observations and m censored observations.

Type III (random) censoring occurs when the component under study is removed from the experiment because another event has occurred. Two distributions are generated: one for uncensored survival times T and another for the censor-

$$\text{Observed} = \begin{cases} 1, & \text{if } T \leq t_c \\ 0, & \text{if } T > t_c \end{cases}$$

where T is the survival time of the observation.

$$\text{Observed} = \begin{cases} T_i, & \text{if } i \leq n - m \\ T_{(n-m+1)}, & \text{if } i > n - m \end{cases}$$

ing mechanism C . An event is censored if the uncensored survival time exceeds the censoring time.

$$\text{Observed} = \begin{cases} 1, & \text{if } T \leq C \\ 0, & \text{if } T > C \end{cases}$$

Sample Generation Method: Controlling censorship percentage θ

Type I, to determine the desired censorship percentage in this censorship scheme θ for $T \sim F_T$ random variables, simply use the quantile function to calculate threshold t_c .

Type II, to determine the desired censorship percentage in this censorship scheme θ for random variables $T \sim F_T$, the data is ordered in ascending order for right censoring or descending order for left censoring. The censoring starts from the m -th order statistic, where $m = \lfloor n \cdot \theta \rfloor$.

Type III, to find the desired censorship percentage in the random censorship scheme $\theta(\lambda)$, we assumed independence between the random variable $C \sim F_C(\cdot | \lambda)$ and $T \sim F_T(\cdot | \alpha)$, α fixed. Case of right censored data, we have the following situation,

$$(C, 0) \iff T = C \iff C < T.$$

Hence, by definition, $P(C < T)$, for $C \perp T$, is given by

$$\begin{aligned} \theta &= P(C < T) = \int_0^\infty P(C < t | T = t) f_T(t) dt \\ &= \int_0^\infty F_C(x | \lambda) f_T(t) dt = \mathbb{E}_T[F_C(T | \lambda)] \\ \theta &\approx \frac{1}{n} \sum_{i=1}^n F_C(t_i | \lambda), \text{ with } t_i \stackrel{iid}{\sim} F_T(\cdot | \alpha). \end{aligned}$$

Then numerically solve the following equation:

$$0 = \theta - \frac{1}{n} \sum_{i=1}^n F_C(t_i | \lambda) = G(\lambda).$$

CRAN Package: "rcens" [2]

- `rcensT1(rdistrX, qdistrX, param_X, n, theta, right)` to generate censorship sample type I.
- `rcensT2(rdistrX, param_X, n, theta, right)` to generate censorship sample type II.
- `rcensT3(rdistrX, param_X, rdistrC, param_C, n, theta, right)` to generate censorship sample type III.
- `rcensI(rdistrX, param_X, rdistrC, param_C, n, theta)` to generate interval censoring sample.

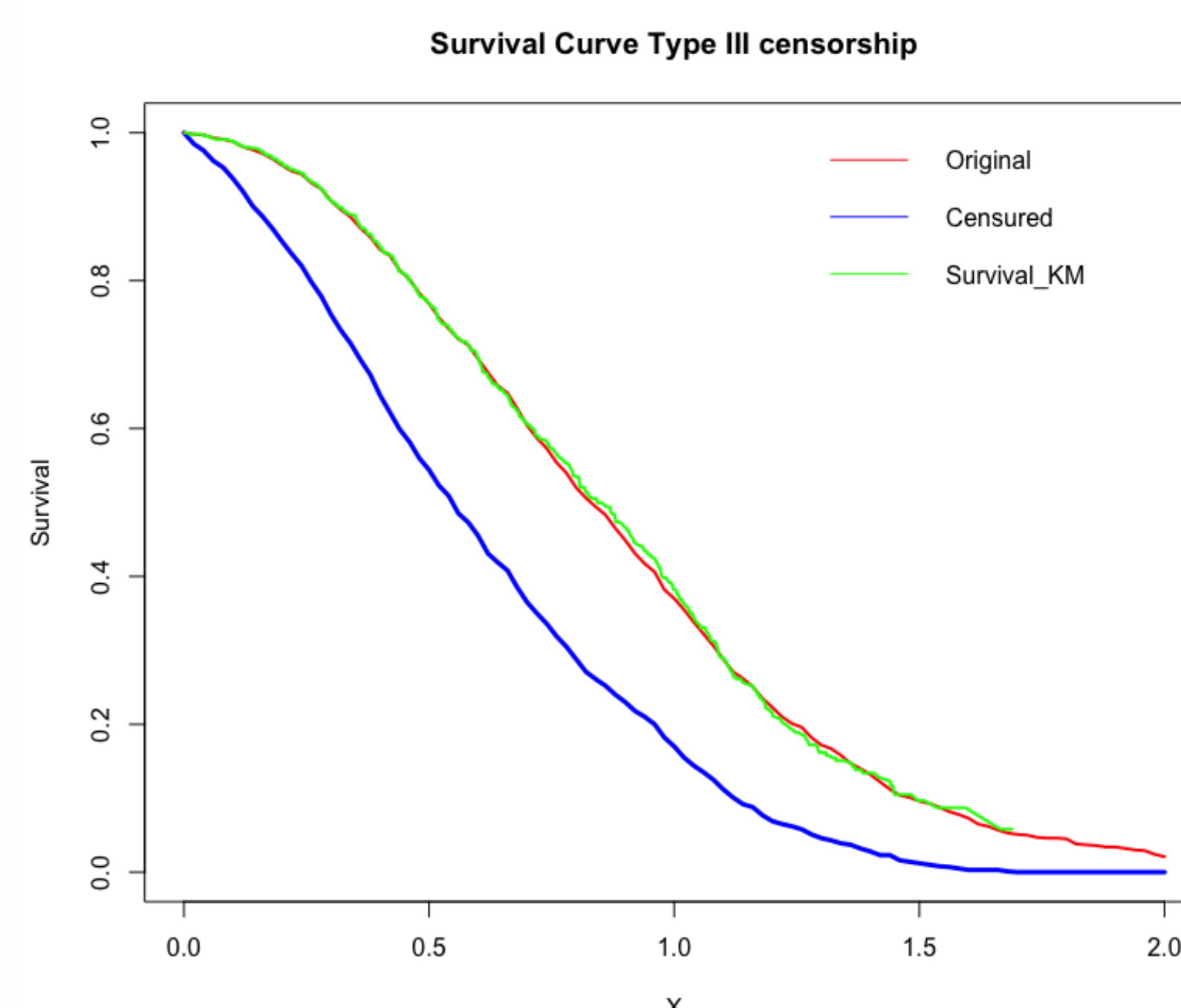


Figure 1: Type III right censoring with $T \sim \text{Weibull}(2, 1)$, $C \sim \text{Unif}(0, \lambda)$, and a censoring percentage of $\theta = 0.5$.

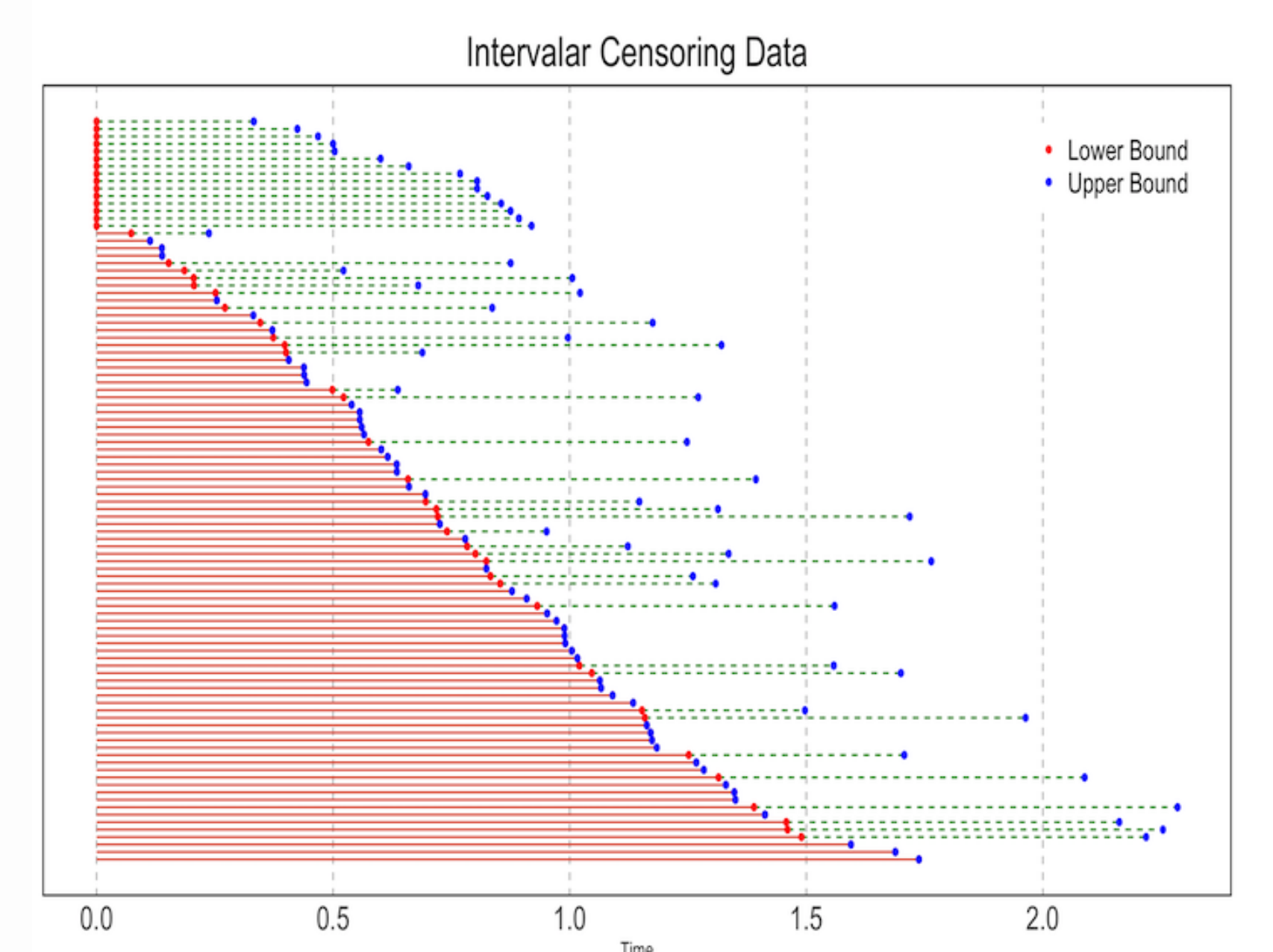


Figure 2: Interval censoring with $T \sim \text{Weibull}(2, 1)$, time between visits $C \sim \text{Unif}(0, 1)$, and a censoring percentage of $\theta = 0.5$.

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